SG No. 226 O_{h}^{6} $Fm\bar{3}c$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `432`

No.	position	mapping
1	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`
2	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `8b`, site symmetry: `m-3.`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,25,26,27,28,29,30,31,32,33,34,35,36]`
2	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 3: Wyckoff site: `24c`, site symmetry: `-4m.2`

No.	position	mapping
1	$\left[\frac{1}{4},\,0,\,0\right]$	`[1,4,18,19,26,27,41,44]`
2	$\left[\frac{3}{4},\,0,\,0\right]$	`[2,3,17,20,25,28,42,43]`
3	$\left[0,\,\frac{1}{4},\,0\right]$	`[5,8,14,15,30,31,37,40]`
4	$\left[0,\,\frac{3}{4},\,0\right]$	`[6,7,13,16,29,32,38,39]`
5	$\left[0,\,0,\,\frac{1}{4}\right]$	`[9,12,21,24,34,35,46,47]`
6	$\left[0,\,0,\,\frac{3}{4}\right]$	`[10,11,22,23,33,36,45,48]`

Table 4: Wyckoff site: `24d`, site symmetry: `4/m..`

No.	position	mapping
1	$\left[0,\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,4,17,20,25,28,41,44]`
2	$\left[0,\,\frac{3}{4},\,\frac{1}{4}\right]$	`[2,3,18,19,26,27,42,43]`
3	$\left[\frac{1}{4},\,0,\,\frac{1}{4}\right]$	`[5,8,13,16,29,32,37,40]`
4	$\left[\frac{1}{4},\,0,\,\frac{3}{4}\right]$	`[6,7,14,15,30,31,38,39]`
5	$\left[\frac{1}{4},\,\frac{1}{4},\,0\right]$	`[9,12,22,23,33,36,46,47]`
6	$\left[\frac{3}{4},\,\frac{1}{4},\,0\right]$	`[10,11,21,24,34,35,45,48]`

Table 5: Wyckoff site: `48e`, site symmetry: `mm2..`

No.	position	mapping
1	$[x,\,0,\,0]$	`[1,4,26,27]`
2	$[-x,\,0,\,0]$	`[2,3,25,28]`
3	$[0,\,x,\,0]$	`[5,8,30,31]`
4	$[0,\,-x,\,0]$	`[6,7,29,32]`
5	$[0,\,0,\,x]$	`[9,12,34,35]`
6	$[0,\,0,\,-x]$	`[10,11,33,36]`
7	$\left[\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}\right]$	`[13,16,38,39]`

continued ...

Table 5

No.	position	mapping
8	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[14,15,37,40]
9	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,20,42,43]
10	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,19,41,44]
11	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[21,24,46,47]
12	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[22,23,45,48]

Table 6: Wyckoff site: 48f, site symmetry: 4..

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,4,17,20]
2	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,3,18,19]
3	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,8,13,16]
4	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[6,7,14,15]
5	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9,12,22,23]
6	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[10,11,21,24]
7	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[25,28,41,44]
8	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[26,27,42,43]
9	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[29,32,37,40]
10	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[30,31,38,39]
11	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[33,36,46,47]
12	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[34,35,45,48]

Table 7: Wyckoff site: 64g, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,9]
2	$[-x, -x, x]$	[2,7,12]
3	$[-x, x, -x]$	[3,8,10]
4	$[x, -x, -x]$	[4,6,11]
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[13,17,21]
6	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[14,19,24]
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[15,20,22]
8	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[16,18,23]
9	$[-x, -x, -x]$	[25,29,33]
10	$[x, x, -x]$	[26,31,36]
11	$[x, -x, x]$	[27,32,34]
12	$[-x, x, x]$	[28,30,35]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[37,41,45]
14	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[38,43,48]
15	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[39,44,46]
16	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[40,42,47]

Table 8: Wyckoff site: `96h`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ y\right]$	`[1,18]`
2	$\left[\frac{3}{4},\ -y,\ y\right]$	`[2,20]`
3	$\left[\frac{3}{4},\ y,\ -y\right]$	`[3,17]`
4	$\left[\frac{1}{4},\ -y,\ -y\right]$	`[4,19]`
5	$\left[y,\ \frac{1}{4},\ y\right]$	`[5,15]`
6	$\left[y,\ \frac{3}{4},\ -y\right]$	`[6,13]`
7	$\left[-y,\ \frac{3}{4},\ y\right]$	`[7,16]`
8	$\left[-y,\ \frac{1}{4},\ -y\right]$	`[8,14]`
9	$\left[y,\ y,\ \frac{1}{4}\right]$	`[9,21]`
10	$\left[-y,\ y,\ \frac{3}{4}\right]$	`[10,23]`
11	$\left[y,\ -y,\ \frac{3}{4}\right]$	`[11,22]`
12	$\left[-y,\ -y,\ \frac{1}{4}\right]$	`[12,24]`
13	$\left[\frac{3}{4},\ -y,\ -y\right]$	`[25,42]`
14	$\left[\frac{1}{4},\ y,\ -y\right]$	`[26,44]`
15	$\left[\frac{1}{4},\ -y,\ y\right]$	`[27,41]`
16	$\left[\frac{3}{4},\ y,\ y\right]$	`[28,43]`
17	$\left[-y,\ \frac{3}{4},\ -y\right]$	`[29,39]`
18	$\left[-y,\ \frac{1}{4},\ y\right]$	`[30,37]`
19	$\left[y,\ \frac{1}{4},\ -y\right]$	`[31,40]`
20	$\left[y,\ \frac{3}{4},\ y\right]$	`[32,38]`
21	$\left[-y,\ -y,\ \frac{3}{4}\right]$	`[33,45]`
22	$\left[y,\ -y,\ \frac{1}{4}\right]$	`[34,47]`
23	$\left[-y,\ y,\ \frac{1}{4}\right]$	`[35,46]`
24	$\left[y,\ y,\ \frac{3}{4}\right]$	`[36,48]`

Table 9: Wyckoff site: `96i`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,28]`
2	$[0,\ -y,\ z]$	`[2,27]`
3	$[0,\ y,\ -z]$	`[3,26]`
4	$[0,\ -y,\ -z]$	`[4,25]`
5	$[z,\ 0,\ y]$	`[5,32]`
6	$[z,\ 0,\ -y]$	`[6,31]`
7	$[-z,\ 0,\ y]$	`[7,30]`
8	$[-z,\ 0,\ -y]$	`[8,29]`
9	$[y,\ z,\ 0]$	`[9,36]`
10	$[-y,\ z,\ 0]$	`[10,35]`
11	$[y,\ -z,\ 0]$	`[11,34]`
12	$[-y,\ -z,\ 0]$	`[12,33]`
13	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,40]`
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,39]`
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,38]`
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[16,37]`
17	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[17,44]`

continued ...

Table 9

No.	position	mapping
18	$\left[\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[18,43]
19	$\left[\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[19,42]
20	$\left[\frac{1}{2},\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[20,41]
21	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[21,48]
22	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[22,47]
23	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[23,46]
24	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[24,45]

Table 10: Wyckoff site: 192j, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$[x,\, -y,\, -z]$	[4]
5	$[z,\, x,\, y]$	[5]
6	$[z,\, -x,\, -y]$	[6]
7	$[-z,\, -x,\, y]$	[7]
8	$[-z,\, x,\, -y]$	[8]
9	$[y,\, z,\, x]$	[9]
10	$[-y,\, z,\, -x]$	[10]
11	$[y,\, -z,\, -x]$	[11]
12	$[-y,\, -z,\, x]$	[12]
13	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[14]
15	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[17]
18	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[18]
19	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[19]
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[20]
21	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[21]
22	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[22]
23	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[23]
24	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[24]
25	$[-x,\, -y,\, -z]$	[25]
26	$[x,\, y,\, -z]$	[26]
27	$[x,\, -y,\, z]$	[27]
28	$[-x,\, y,\, z]$	[28]
29	$[-z,\, -x,\, -y]$	[29]
30	$[-z,\, x,\, y]$	[30]
31	$[z,\, x,\, -y]$	[31]
32	$[z,\, -x,\, y]$	[32]
33	$[-y,\, -z,\, -x]$	[33]
34	$[y,\, -z,\, x]$	[34]

continued ...

Table 10

No.	position	mapping
35	$[-y,\, z,\, x]$	[35]
36	$[y,\, z,\, -x]$	[36]
37	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[37]
38	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[38]
39	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[39]
40	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[40]
41	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[41]
42	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[42]
43	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[43]
44	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[44]
45	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[45]
46	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[46]
47	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[47]
48	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[48]